

SEMESTER-V

COURSE 11: BUSINESS INTELLIGENCE TOOLS

Theory

Credits: 3

3 hrs/week

Course Objectives:

1. Introduce foundational concepts of Business Intelligence (BI) and Decision Support Systems (DSS), including their scope, evolution, and organizational relevance.
2. Familiarize students with leading BI tools such as Power BI and Tableau, highlighting their ecosystems, interfaces, and comparative strengths.
3. Develop skills in data preparation and transformation, using Power Query and Tableau's data connection features to clean and model datasets.
4. Enable effective data visualization and storytelling, leveraging charts, dashboards, and advanced features to communicate insights.
5. Equip learners with data modeling techniques, including dimensional modeling, relationships, joins, and governance principles for robust BI solutions.

Course Outcomes:

At the end of the course, the students will be able to:

1. Differentiate between BI, Data Analytics, and Data Science, and explain the BI lifecycle and its applications across functional domains.
2. Use Power BI and Tableau to prepare, transform, and visualize data, applying basic DAX functions and calculated fields for analysis.
3. Design and implement dimensional data models, including star and snowflake schemas, and apply relationships and joins in BI tools.
4. Create interactive dashboards and visualizations, incorporating parameters, slicers, filters, and drilldowns to enhance decision-making.
5. Build and publish complete BI dashboards, and effectively communicate business insights through storytelling and visualization best practices.

Unit-I: Introduction to Business Intelligence and Decision Support Systems

Business Intelligence: Definition, Scope, and Evolution, Business IntelligenceI vs. Data Analytics vs. Data Science, BI Lifecycle, **Applications of BI in Functional Domains:** Finance, HR, Marketing, Retail, Education, Healthcare, etc., BI Maturity Models &

Organizational Readiness to BI adoption Decision Support Systems (DSS): Concepts, Components, and Architecture.

BI Tools Overview: Power BI, Tableau, and other tools, Comparison and suitability of BI Tools.

Case Study: Retail Chain's BI Strategy to Optimize Inventory

Unit-II: Data Preparation and Visualization with Power BI

Introduction to Power BI, Power BI Ecosystem: Desktop, Service, Mobile; Power BI Interface; Data Sources: Excel, CSV, SQL Server, Web APIs, Power Query: Data Preparation, Cleaning & Transformation - Connect, transform, and model a dataset; Basic DAX Functions: SUM, COUNT, AVERAGE, CALCULATE, IF; Creating Simple Visualizations: Charts, Tables, Cards; Sharing Reports via Power BI Service.

Case Study: Student performance analysis in Higher Education , Analyze Finance Dataset

Unit-III: Data Preparation, Visualization and Storytelling with Tableau

Introduction to Tableau; Characteristics of Tableau; Tableau Architecture and components - Tableau Public, Desktop, Reader, Online, Server; Tableau Interface: Shelves, Marks Card, Views; Tableau extensions, Data Connection and Preparation: Cleaning, Pivoting, Filtering; Calculated Fields and LOD Expressions; Basic Visualizations: Bar, Line, Tree, Geo Maps, Scatter Plots; Storytelling with Tableau, Creating a Tableau story.

Case Study: HR Analytics

Unit-IV: Data Modeling and Relationships in BI Tools

Dimensional Modeling: Dimension, Dimension table, fact, fact table, schema, Star and Snowflake Schemas

Power BI: Relationships, Cardinality, Cross-filtering; Tableau: Joins (Inner, Left, Full), Blending; Data Governance: Metadata, Hierarchies, Quality; Data Model Design Best Practices - Design and implement data model in Power BI and Tableau; HR or Retail data model design and insights

Case Study: Retail BI for Sales Optimization

Unit-V: Dashboard Design and Business Insights

Introduction to Dashboard, when to use dashboards, Dashboard components, Principles of Effective Visualization & Dashboarding, Advanced Visualizations: Parameters, Slicers, Filters, Drilldowns, Graphs and Maps, Dashboard Design: Layout, Alignment, Accessibility

Publishing Dashboards: Power BI Service, Tableau Public; Storytelling and Insight Communication, Build a complete BI dashboard using either tool.

Case Study: Business decision-making scenario (e.g., Sales Forecasting, Budgeting)

Text Books:

1. Decision Support and Business Intelligence Systems (9th ed.). Turban, E., Sharda, R., & Delen, D. (2014), Pearson Education.
2. Learning Tableau 2022: Create effective data visualizations, build interactive dashboards, and transform your data into insights (6th Edition), Milligan, J. N. (2022), Packt Publishing.
3. Expert Data Modeling with Power BI: Enrich and optimize your data models for reporting and business needs (2nd Edition). Bakhshi, S. (2023), Packt Publishing.

Reference Books:

1. Visual Analytics with Tableau, Loth, A. (2019), Addison-Wesley Professional.
2. The Definitive Guide to DAX: Business intelligence for Microsoft Power BI, SQL Server Analysis Services, and Excel (2nd Edition), Russo, M., & Ferrari, A. (2020), Microsoft Press.

Online Resources:

1. [Decision Support Systems Courses – Class Central](#)
2. [Advanced Business Decision Support Systems – NPTEL \(IIT Kanpur\)](#)
3. [Power BI Learning Paths – Microsoft Learn](#)
4. [Power BI Courses – Coursera](#)
5. [Tableau Learning Hub – Official Site](#)
6. [Free Tableau Course – Simplilearn](#)
7. [Dashboard Design Concepts – DataCamp](#)
8. [Business Intelligence with Power BI – Swayam](#)

Activities:

1. **Differentiate BI, Data Analytics, and Data Science + BI Lifecycle**

Activity: Case Study Analysis: Provide students with a business scenario and ask them to identify how BI, analytics, and data science each contribute. Then, have them map the BI lifecycle stages to that scenario and explain its impact across departments (e.g., finance, marketing).

Evaluation Method: Evaluate on a 10-point scale based on Conceptual differentiation clarity (30%), BI lifecycle accuracy (30%), Application to domains (20%) and Written summary or presentation quality (20%)

2. Prepare, Transform & Visualize Data in Power BI/Tableau (with DAX & Calculated Fields)

Activity: Hands-on Data Challenge: Supply students with a raw dataset (e.g. sales or customer data). Task them with cleaning, transforming, and building visuals in Power BI/Tableau using basic DAX and calculated fields (e.g. profit margins, year-over-year growth).

Evaluation Method: Evaluate for 10-points based on Effective transformation workflow (25%), Correct DAX/formula usage (25%), Visualization relevance & clarity (25%) and Documentation of process (25%)

3. Dimensional Modelling with Star/Snowflake Schemas + Joins/Relationships

Activity: Model Building Lab: Students design a star or snowflake schema based on a retail or HR database. Then, implement the schema in Power BI or Tableau and create relationships between tables to ensure correct joins and data flow.

Evaluation Method: Evaluate on a 10-point scale based on Schema design accuracy (30%), Appropriate schema selection (star vs. snowflake) (20%), Implementation of joins/relationships (30%) and Functional model validation (20%)

4. Create Interactive Dashboards with Advanced Features

Activity: Dashboard Design Workshop: Students build an interactive dashboard using parameters, slicers, filters, and drilldowns to simulate real-time decision-making (e.g., tracking regional product sales or employee performance across departments).

Evaluation Method: Evaluate on a 10-point scale on the basis of Integration of interactive features (30%), Usability and navigation experience (30%), Data-driven insights extracted (20%) and Design polish and layout consistency (20%).

5. Build and Publish BI Dashboards + Business Storytelling

Activity: BI Capstone Project: Students design and publish a complete dashboard solving a real or simulated business problem (e.g. customer churn, supply chain bottlenecks). Include visual storytelling techniques-titles, annotations, color themes, and narratives.

Evaluation Method: Evaluate on a 10-point scale on the basis of Clarity of business insights communicated (30%), Storytelling elements (25%), Dashboard completeness and polish (25%) and Peer or instructor presentation (20%)

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COURSE 11: BUSINESS INTELLIGENCE TOOLS

Practical

Credits: 1

2 hrs/week

List of Experiments:

1. Exploring BI Tools - Power BI vs Tableau.
2. Create a simple retail dashboard using both tools.
3. **Connecting to Different Data Sources in Power BI**
 - a. Learn to connect Excel, CSV, and Web data.
 - b. Load a dataset from different formats.
 - c. Demonstrate dataset loading and view schema.
4. **Data Cleaning and Transformation using Power Query**
 - a. Apply Power Query for cleaning.
 - b. Remove duplicates, fill nulls, filter rows.
 - c. Submit Power Query steps and cleaned dataset.
5. **Prepare and clean a higher education student performance dataset using Power Query and visualize key academic metrics.**
 - a. Connect to a dataset and perform data cleaning, transformation, and reshaping using Power Query.
 - b. Visualize academic performance indicators such as GPA trends, pass rates, or subject-wise scores.
 - c. Upload file containing the cleaned dataset and relevant academic metric visualizations.
6. **Implementing DAX Functions**
 - a. Use DAX to perform basic calculations.
 - b. Create calculated columns with SUM, AVERAGE, COUNT, CALCULATE, IF.
 - c. Submit DAX expressions with visual output.
7. **Creating Basic Visualizations in Power BI**
 - a. Develop simple charts and cards.
 - b. Use sales or HR data to create bar, pie charts, and KPIs.
 - c. Display Dashboard showing at least 3 chart types.
8. **Tableau Basics and Connecting to Data**
 - a. Connect and explore data in Tableau Public.

- b. Load a student performance or any other dataset and preview data.
 - c. Upload dataset and share working link.
- 9. **Visualize employee turnover patterns using Tableau and apply LOD expressions to uncover retention drivers.**
 - a. Import and clean HR data for turnover analysis in Tableau.
 - b. Apply LOD expressions to identify patterns across departments, roles, and tenure.
 - c. Upload cleaned dataset and interactive visualizations highlighting retention insights.
- 10. **Data Cleaning, Pivoting & Filtering in Tableau**
 - a. Prepare data inside Tableau.
 - b. Pivot columns, apply filters, and rename headers.
 - c. Upload cleaned worksheet.
- 11. **Creating Visualizations in Tableau**
 - a. Use Marks Card, Shelves, and Views for visualization.
 - b. Build bar chart, map view, scatter plot.
 - c. One dashboard with 3 visuals.
- 12. **Creating a Tableau Story - (Eg HR or Student Performance or any other)**
 - a. Build a narrative with visualizations.
 - b. Combine charts into a Tableau story with captions.
 - c. Publish and present story with link.
- 13. **Designing Data Models in Power BI**
 - a. Create dimensional models (Star/Snowflake schema).
 - b. Use a retail dataset with multiple tables and define relationships.
 - c. Submit relationship diagram and schema explanation.
- 14. **Joins and Blending in Tableau**
 - a. Implement data joins and blending techniques.
 - b. Combine multiple data tables using inner/left/full joins.
 - c. Demonstrate join types with visuals.
- 15. **Dashboard with Drill-downs, Filters, and Slicers**
 - a. Use interactivity in dashboards.
 - b. Build a multi-level dashboard in Power BI with drill-through.
 - c. Submit .pbix file and link to published dashboard.

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COURSE 12 A: MACHINE LEARNING

Theory

Credits: 3

3 hrs/week

Course Objectives:

1. Understand fundamental concepts, types, and applications of machine learning.
2. Develop, evaluate, and optimize machine learning models through preprocessing, training, and feature engineering techniques.
3. Apply supervised and unsupervised learning algorithms to real-world problems using appropriate tools and methods.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. Describe various machine learning paradigms, data types, and the overall structure of a machine learning pipeline.
2. Perform data preprocessing, feature engineering, and evaluate models using appropriate metrics.
3. Implement and analyze supervised learning algorithms for regression and classification tasks.
4. Apply unsupervised learning techniques for clustering and identify suitable machine learning approaches for specific application domains

Unit 1. Introduction to Machine Learning:

Introduction to Machine Learning: Types of human learning, What is machine learning?, Types of machine learning: supervised, unsupervised, semi-supervised and reinforcement learning, machine learning activities, applications of machine learning. Types of data in machine learning, Structure of data

Unit 2. Model Preparation, Evaluation and feature engineering:

Data pre-processing, Model selection and training (for supervised learning), Model representation and interpretability, Evaluating machine learning algorithms and performance enhancement of models. What is feature engineering?, Feature transformation, Feature subset selection. Principal component analysis.

Unit 3. Supervised Learning-Regression:

Regression: Introduction of regression, Regression algorithms: Simple linear regression, Multiple linear regression, Polynomial regression model, Logistic regression, Maximum likelihood estimation.

Unit 4. Supervised Learning- Classification:

Introduction of supervised learning, Classification model and learning steps, Classification algorithms: Naïve Bayes classifier, k-Nearest Neighbour (kNN), Decision tree, Support vector machines, Random Forest.

Unit 5. Unsupervised Learning:

Introduction of unsupervised learning, Unsupervised vs supervised learning, Application of unsupervised learning, Clustering and its types, Partitioning method: k-Means and KMedoids, Hierarchical clustering, Density-based methods - DBSCAN.

Case-study of ML applications: Image recognition, speech recognition, Email spam filtering, Online fraud detection and other.

Textbooks:

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, Fourth Edition, 2020.
2. Machine Learning: Theory and Practice, M N Murthy, V.S Ananthanarayana, Universities press
3. Machine Learning, S. Sridhar, M. Vijayalakshmi, Second Edition, Oxford University Press

Reference Books:

1. Machine Learning: An Algorithmic Perspective, Second Edition, Stephen Marsland, CRC Press, 2014
2. Machine Learning, Tom Mitchell, McGraw Hill, 3rd Edition.
3. Python Machine Learning, Sebastain Raschka, Vahid Mirjalili , Packt publishing 3rd Edition, 2019.

Activities:

Outcome: Describe various machine learning paradigms, data types, and the overall structure of a machine learning pipeline.

Activity: Prepare a detailed comparative report/chart explaining supervised, unsupervised, and reinforcement learning paradigms, data types, and step-by-step machine learning pipeline stages.

Evaluation Method: Rubric-based assessment evaluating completeness, clarity, correctness, and presentation quality - scored on a 10-point scale.

Outcome: Perform data preprocessing, feature engineering, and evaluate models using appropriate metrics.

Activity: Conduct a hands-on lab exercise using a real dataset to perform data cleaning, normalization, feature extraction/selection, and evaluate model performance using metrics like accuracy, precision, recall, and F1-score.

Evaluation Method: Practical assessment including code correctness, applied techniques, and interpretation of evaluation metrics; assessed with a rubric out of 10.

Outcome: Implement and analyze supervised learning algorithms for regression and classification tasks.

Activity: Implement at least two supervised learning algorithms (e.g., Linear Regression and Decision Trees) to solve prediction tasks, followed by comparative analysis of their performance on test datasets.

Evaluation Method: Code and report evaluation focusing on implementation accuracy, performance comparison, and analysis depth; scored on a 10-point rubric.

Outcome: Apply unsupervised learning techniques for clustering and identify suitable machine learning approaches for specific application domains.

Activity: Perform clustering (e.g., K-Means, Hierarchical) on a given dataset and prepare a case study selecting and justifying machine learning methods suited for different application scenarios.

Evaluation Method: Lab practical combined with a written case study; assessed for correct algorithm application, cluster interpretation, and justification of approach - evaluated on a 10-point rubric.

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COURSE 12 A: MACHINE LEARNING

Practical

Credits: 1

2 hrs/week

Lab Experiments:

1. Write a python program to import and export data using Pandas library functions.
2. Demonstrate various data pre-processing techniques for a given dataset
3. Implement Dimensionality reduction using the Principal Component Analysis (PCA) method.
4. Write a Python program to demonstrate various Data Visualization Techniques.
5. Implement MLE on a Dataset
6. Implement Simple and Multiple Linear Regression Models.
7. Develop Logistic Regression Model for a given dataset.
8. Develop Decision Tree Classification model for a given dataset and use it to classify a new sample.
9. Implement Naïve Bayes Classification in Python.
10. Develop K-Means for a Given Dataset
11. Build KNN Classification model for a given dataset.
12. Develop DBSCAN on a given Dataset

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COURSE 12 B: BIG DATA TECHNOLOGIES

Theory

Credits: 3

3 hrs/week

Course Objectives:

1. Introduce students to the concepts, characteristics, and challenges of Big Data.
2. Familiarize students with the Hadoop ecosystem and its core components (HDFS, YARN, MapReduce).
3. Develop practical knowledge of distributed storage and parallel processing in Hadoop.
4. Provide hands-on exposure to data ingestion tools (Sqoop, Flume) and serialization techniques.
5. Enable students to explore NoSQL databases (HBase), coordination services (ZooKeeper), and Hadoop–Spark integration for large-scale data analysis.

Course Outcomes:

At the end of this course, students will be able to:

1. Explain Big Data concepts and challenges along with the role of the Hadoop ecosystem.
2. Demonstrate understanding of HDFS and YARN architectures and their functions in distributed data management.
3. Apply MapReduce and high-level tools (Hive, Pig, Spark) to process and analyze large datasets.
4. Design and implement data ingestion workflows using Sqoop, Flume, and serialization formats like Avro and Parquet.
5. Utilize NoSQL databases and ecosystem enhancements such as HBase, ZooKeeper, and Hadoop–Spark integration for scalable big data solutions.

Unit 1. Foundations of Big Data & Hadoop Ecosystem

Introduction to Big Data: characteristics (volume, variety, velocity, veracity, value), Hadoop Ecosystem Overview: HDFS, MapReduce, YARN, Hadoop Common, Hadoop architecture and use cases

Unit 2. Hadoop Distributed File System (HDFS) & YARN:

Deep dive into HDFS architecture: blocks, NameNode, DataNodes, HDFS file operations, fault tolerance, replication

YARN architecture: ResourceManager, NodeManager, application scheduling

Unit 3. MapReduce & High-Level Tools

MapReduce programming model: map, shuffle, reduce phases, Writing MapReduce applications in Hadoop

High-level abstractions: Hive, Pig, Crunch, and introduction to Spark integration

Unit 4. Data Ingestion & Serialization

Data ingestion pipelines: Sqoop (for RDBMS), Flume (streaming), Data formats & serialization: Avro, Parquet, SequenceFile, Practical ingestion workflows-batch and streaming

Unit 5. NoSQL & Ecosystem Enhancements

Overview of NoSQL within Hadoop ecosystem: HBase, Configuration and usage of ZooKeeper for coordination, Hadoop integration with Spark for data processing

Textbooks

1. Hadoop: The Definitive Guide, Tom White, 4th Edition, O'Reilly
Free Resource available at [piazza-resources.s3.amazonaws.comO'Reilly Media](http://piazza-resources.s3.amazonaws.com/O'Reilly Media)
2. Learning Spark, 2nd Edition, Jules S. Damji, Brooke Wenig, Tathagata Das, Denny Lee, O'Reilly

Reference Books

3. BIG DATA, Black Book TM, DreamTech Press, 2016 Edition.
4. BIG DATA and ANALYTICS, Seema Acharya, SubhasniChellappan , Wiley publications, 2016

Activities:

Outcome: Explain Big Data concepts and Hadoop ecosystem

Activity: Short seminar / presentation on Big Data applications

Evaluation Method: Oral presentation + concept quiz

Outcome: Demonstrate HDFS and YARN architectures

Activity: Hands-on lab to configure HDFS & analyze NameNode/DataNode logs

Evaluation Method: Lab performance + viva

Outcome: Apply MapReduce and high-level tools

Activity: Mini-project implementing MapReduce job and Hive queries

Evaluation Method: Project report + execution demo

Outcome: Design and implement data ingestion workflows

Activity: Lab task to ingest data using Sqoop/Flume and serialize with Avro/Parquet

Evaluation Method: Lab record + output validation

Outcome: Utilize NoSQL and ecosystem enhancements

Activity: Case study on HBase–Spark integration with example dataset

Evaluation Method: Case study report + written test

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COURSE 12 B: BIG DATA TECHNOLOGIES

Practical	Credits: 1	2 hrs/week
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1. Installation & setup of Hadoop single-node cluster
 2. Explore Hadoop directory structure and basic commands (hadoop fs operations)
 3. Demonstration of Hadoop architecture components (HDFS, YARN, MapReduce) using sample logs
 4. Store and retrieve large files from HDFS (block distribution, replication factor demo)
 5. Simulate NameNode/DataNode failure and observe fault tolerance & recovery
 6. Configure YARN and run sample applications, observe ResourceManager and NodeManager roles
 7. Write a simple MapReduce program for word count
 8. Develop a MapReduce job for inverted index creation
 9. Perform data analysis using Pig Latin scripts
 10. Execute Hive queries for structured data analysis (tables, partitions)
 11. Import data from RDBMS into Hadoop using Sqoop
 12. Capture and store log/streaming data using Flume
 13. Serialize and store datasets in Avro and Parquet formats
 14. Build an end-to-end ingestion workflow combining batch (Sqoop) and streaming (Flume)
 15. Create and manage tables in HBase (CRUD operations)
 16. Demonstrate coordination with ZooKeeper
 17. Process HBase datasets using Spark integration with Hadoop

SEMESTER-V

COURSE 13 A: ARTIFICIAL INTELLIGENCE

Theory

Credits: 3

3 hrs/week

Course Objectives

1. Understand the fundamental concepts, history, types, and applications of Artificial Intelligence.
2. Develop problem-solving skills using state-space representations and search strategies for AI applications.
3. Apply informed and advanced search techniques including heuristics, local search, genetic algorithms, and constraint satisfaction problems.
4. Learn knowledge representation methods and reasoning techniques using propositional and first-order logic for intelligent agents.
5. Explore expert systems, probabilistic reasoning, fuzzy logic, and emerging AI technologies including NLP, robotics, and ethical considerations.

Course Outcomes

At the end of the course, students will be able to:

1. Explain the concepts, scope, types of AI, and structure of intelligent agents and their environments.
2. Formulate problems using state-space representation and solve them using uninformed search strategies like BFS, DFS, and Uniform Cost Search.
3. Apply informed search, local search, genetic algorithms, and constraint satisfaction techniques to solve complex AI problems.
4. Represent knowledge using propositional and first-order logic, perform reasoning, and design knowledge-based agents.
5. Design simple expert systems, implement probabilistic reasoning and fuzzy logic, and demonstrate awareness of emerging AI technologies and ethical issues.

Unit 1. Introduction to Artificial Intelligence & Intelligent Agents:

Definition and scope of AI, history and evolution of AI, Turing Test, Applications of AI in real world.

Types of AI: Weak AI vs Strong AI, Narrow AI vs General AI. Intelligent Agents: Structure of agents, Rationality, Agent types. Environments: Deterministic vs Stochastic, Static vs Dynamic, Discrete vs Continuous. PEAS representation (Performance measure, Environment, Actuators, Sensors).

Unit 2. Problem Solving: State Space & Uninformed Search:

State space representation: Components (State, Actions, Goal test, Path cost). Problem formulation and examples (8-puzzle, water jug, vacuum cleaner world).

Uninformed search strategies: Breadth First Search (BFS), Depth First Search (DFS), Uniform Cost Search- Properties: Completeness, Optimality, Time & Space complexity.

Unit 3. Informed & Advanced Search Strategies:

Informed search strategies: Heuristics (concept, admissibility, consistency), Greedy Best First Search, A* Algorithm

Local Search: Hill Climbing, Simulated Annealing. Genetic Algorithms

Constraint Satisfaction Problems (CSP): Definition, Backtracking search.

Unit 4. Knowledge Representation & Reasoning:

Knowledge Representation: Issues, Approaches.

Propositional Logic: Syntax, Semantics, Truth tables, Inference rules.

First Order Logic (FOL): Syntax, Semantics, Quantifiers, Substitution, Unification.

Inference in Logic: Forward Chaining, Backward Chaining, Resolution.

Knowledge-based agents.

Unit 5. Expert Systems, Probabilistic & Emerging AI:

Expert Systems: Architecture, Knowledge base, Inference engine, Explanation facility.

Probabilistic Reasoning: Bayes Theorem, Bayesian Belief Networks (concepts & examples)

Fuzzy Logic and Uncertainty handling.

Emerging topics: NLP basics, Robotics, AI Ethics & societal impact.

Textbooks:

1. Artificial Intelligence: A Modern Approach, Stuart Russell & Peter Norvig, 4th Edition, Pearson
2. Artificial Intelligence, Elaine Rich & Kevin Knight, 3rd Edition, McGraw-Hill

Reference Books:

1. The Art of Prolog, Leon Sterling & Ehud Shapiro, MIT Press
2. Learn Prolog Now, Patrick Blackburn, Johan Bos, Kristina Striegnitz (Free online book)

Activities:

Outcome: Explain the concepts, scope, types of AI, and structure of intelligent agents and their environments.

Activity: Divide the class into small groups and ask each group to create a poster or digital infographic comparing Weak AI vs Strong AI, and Narrow AI vs General AI, including real-world examples of intelligent agents and their environments.

Evaluation Method: Peer and instructor review rubric (10 points) assessing correctness of concepts, clarity of illustrations, examples provided, and creativity in presentation.

Outcome: Formulate problems using state-space representation and solve them using uninformed search strategies like BFS, DFS, and Uniform Cost Search.

Activity: Provide students with a simple problem like the 8-puzzle or water jug problem. Students model the state space, draw the state tree, and manually perform BFS and DFS traversal to reach the goal state.

Evaluation Method: Submission of state-space diagrams and solution steps, graded on accuracy, completeness, and correct application of search strategies.

Outcome 3: Apply informed search, local search, genetic algorithms, and constraint satisfaction techniques to solve complex AI problems.

Activity: Conduct a mini-coding session in Python where students implement A* search for a maze-solving problem or hill climbing for a simple optimization task. They can also experiment with a genetic algorithm for a simple fitness function problem.

Evaluation Method: Code demonstration and correctness check, including explanation of heuristics, search path, and results.

Outcome: Represent knowledge using propositional and first-order logic, perform reasoning, and design knowledge-based agents.

Activity: Give students a logic puzzle (e.g., Sudoku rules or “who owns which pet” problem). Students write propositional and/or FOL statements, draw inference chains, and perform forward/backward chaining to solve the puzzle.

Evaluation Method: Solution submission and in-class demonstration of inference process, graded on logical correctness, clarity of reasoning, and completeness.

Outcome: Design simple expert systems, implement probabilistic reasoning and fuzzy logic, and demonstrate awareness of emerging AI technologies and ethical issues.

Activity: Ask students to design a rule-based expert system for a small domain (e.g., medical symptom checker), represent a simple Bayesian network for probabilistic reasoning, and prepare a short presentation on AI ethics or NLP applications.

Evaluation Method: Project-based evaluation including expert system rules, Bayesian reasoning correctness, and quality of presentation on emerging AI topics.

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COURSE 13 A: ARTIFICIAL INTELLIGENCE

Practical

Credits: 1

2 hrs/week

SWI-Prolog environment for practice without installation.

1. Write Prolog facts for a family tree.
 - a. Define rules for ancestor/2, sibling/2, cousin/2.
1. Query for ancestors, descendants, and siblings.
2. Implement member/2, append/3, reverse/2, length/2.
3. Write a predicate to find the maximum element of a list.
4. Flatten a nested list into a single-level list.
5. Write Prolog rules to calculate factorial and Fibonacci numbers.
6. Implement GCD of two numbers using recursion.
7. Demonstrate the use of cut (!) and fail predicates.
8. Represent a simple graph using edge/2 predicates.
9. Write a recursive DFS to find a path between two nodes.
10. Implement BFS to find a path between two nodes in a graph.
11. Compare DFS and BFS by finding path lengths.
12. Solve the 8-puzzle or grid problem using Greedy Best-First search and A*.
13. Represent a map with regions and adjacency constraints.
 1. Assign colors to regions using backtracking.
 2. Ensure no adjacent regions share the same color.
14. Place N queens on an N×N chessboard such that no two queens threaten each other.
 1. Generate all possible solutions using backtracking.
15. Encode facts and rules using propositional and first-order logic.
16. Implement forward and backward chaining for simple queries.
17. Build a rule-based expert system (e.g., medical diagnosis or plant disease).
 1. Include knowledge base, inference engine, and explanation facility.
 2. Test the system with sample queries.
18. Write a DCG grammar for simple English sentences.
 1. Parse sentences to generate a syntax tree.
19. Implement a simple deterministic Naïve Bayes calculation for categorical data.

SEMESTER-V

COURSE 13 B: CLOUD COMPUTING FOR DATA SCIENCE

Theory

Credits: 3

3 hrs/week

Course Objectives

1. Introduce the fundamentals of cloud computing and its role in data science.
2. Provide understanding of virtualization, service, and deployment models.
3. Familiarize students with cloud storage, data management, and databases.
4. Expose students to cloud-based big data and machine learning platforms.
5. Train students in building, deploying, and monitoring ML pipelines on the cloud.

Course Outcomes

At the end of this course, students will be able to:

1. Explain cloud computing concepts including service models, deployment models, and virtualization.
2. Demonstrate cloud storage and database services for managing large-scale data.
3. Apply cloud-based platforms to run data science and machine learning workflows.
4. Build and deploy ML models on cloud services using AutoML and managed ML platforms.
5. Evaluate and monitor cloud-deployed solutions with respect to scalability, performance, and cost.

Unit 1. Introduction to Cloud Computing

Definition & Evolution of Cloud Computing, Service-Oriented Architecture (SOA) & Web Services, Utility & Grid Computing concepts, Characteristics of Cloud Computing

Cloud Computing Architecture: Front-end, Back-end, Networking, Delivery Models

Cloud Service Models: SaaS, PaaS, IaaS, Continuous Delivery using PaaS

Unit 2. Virtualization & Deployment Models

Concept & importance of Virtualization, Types of Virtualizations: Application, Network, Desktop, Storage, Server, Data Virtualization

Cloud Deployment Models: Public, Private, Community, Hybrid

Role of Cloud Computing in Data Science, Advantages of Cloud in Machine Learning

Unit 3. Cloud Storage & Data Management

Cloud Storage: Introduction, Benefits, Use Cases (Backup, Archiving, DR, Content Delivery)

Cloud Storage Systems: Block-based, File-based, Object-based storages

Key-Value Databases: Features & limitations

Batch vs. Streaming data for ML pipelines

Cloud Data Warehouses: AWS Redshift, Google BigQuery

Unit 4. Cloud Platforms for Data Science & ML

Machine Learning in the Cloud: Benefits & Limitations, Cloud-based ML Services: AIaaS, GPUaaS

Managed ML Platforms: Overview & advantages, Cloud ML Platforms: AWS SageMaker, Azure ML Studio, Google Cloud AutoML

Unit 5. Training & Deployment of ML on Cloud

Factors for selecting Cloud ML Platforms: ETL/ELT pipeline support, Scale-up/out training, ML frameworks, Pre-tuned services

Steps for Training ML Models in Cloud: Data source identification, Feature engineering, Training, Validation, Deployment, Monitoring, Monitoring & improving cloud-deployed ML models

Case studies & industry applications

Text / Reference Books

1. Handbook of Cloud Computing, Dr. Anand Nayyar, BPB Publications (2019)
2. Cloud Computing: A Practical Approach, Toby Velte, Anthony Velte, Robert C., McGraw Hill
3. Cloud Computing for Data Analysis, Noah Gift, Alfredo Deza, Pragmatic AI Labs
4. Data Science on the Google Cloud Platform, Valliappa Lakshmanan, O'Reilly
5. Machine Learning in the AWS Cloud: Amazon SageMaker, Abhishek Mishra, Wiley

Activities:

Outcome: Explain cloud computing concepts including service models, deployment models, and virtualization.

Activity: Prepare a comparative chart/report on cloud service and deployment models with real-world examples (AWS, Azure, GCP).

Evaluation Method: Report submission & viva (assess clarity, accuracy, and examples used).

Outcome: Demonstrate cloud storage and database services for managing large-scale data.

Activity: Perform lab experiments on block/file/object storage and execute queries in BigQuery / RDS.

Evaluation Method: Lab performance & practical exam (students demonstrate CRUD operations and explain storage use cases).

Outcome: Apply cloud-based platforms to run data science and machine learning workflows.

Activity: Implement a cloud-based ETL pipeline (e.g., using AWS Glue / Dataflow) for preparing a dataset.

Evaluation Method: Lab report + demo evaluation (workflow completeness, correctness of execution).

Outcome: Build and deploy ML models on cloud services using AutoML and managed ML platforms.

Activity: Train and deploy a classification/regression model using AWS SageMaker / Azure ML / Google AutoML.

Evaluation Method: Practical demo + oral viva (assess deployment success, prediction results, and understanding of pipeline).

Outcome: Evaluate and monitor cloud-deployed solutions with respect to scalability, performance, and cost.

Activity: Configure monitoring (e.g., CloudWatch, Stackdriver) for a deployed ML service and analyze resource usage.

Evaluation Method: Mini-project report & presentation (grading based on monitoring setup, analysis quality, cost optimization suggestions).

SEMESTER-V

COURSE 13 B: CLOUD COMPUTING FOR DATA SCIENCE

Practical	Credits: 1	2 hrs/week
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1. Create Virtual Machine using VMware workstation for Windows/Linux
 2. Install & configure WAMP/XAMPP/Apache on the VM and host a sample page
 3. Install and configure a cloud account (AWS/Azure/GCP free tier).
 4. Create and manage storage buckets; upload and access datasets.
 5. Launch an instance and configure Block-based storage (EBS)
 6. Create & configure File-based storage on cloud VM (EFS/Network FS).
 7. Set up Jupyter Notebook/Colab on cloud VM.
 8. Connect to cloud-hosted database services (AWS RDS, BigQuery, Cosmos DB).
 9. Implement a batch ETL pipeline in the cloud.
 10. Launch a SageMaker notebook, attach IAM role and S3 bucket, run sample notebook
 11. Build a classification/regression model using AWS SageMaker / Azure ML Studio / GCP AI Platform.
 12. Implement a simple ETL job: extract (RDS / CSV), transform, load into cloud DW (e.g., Redshift / BigQuery).
 13. Use CloudWatch / Stackdriver to monitor endpoints, set alarms and auto-scale rules.
 14. Use cloud AutoML services for dataset prediction tasks.
 15. Deploy a trained ML model as a REST API endpoint in the cloud.